



Lakshya JEE 2.0 (2024)

Application of Derivative

DPP-01

- An edge of a variable cube is increasing at the rate of 10 cm/sec. How fast the volume of the cube will increase when the edge is 5 cm long
 (1) $750 \text{ cm}^3/\text{sec}$ (2) $75 \text{ cm}^3/\text{sec}$
 (3) $300 \text{ cm}^3/\text{sec}$ (4) $150 \text{ cm}^3/\text{sec}$
- The point on the curve $6y = x^3 + 2$ at which y -co-ordinate is changing 8 times as fast as x -co-ordinate is:
 (1) $[4, 11]$ (2) $[4, -11]$
 (3) $[-4, 11]$ (4) $[-4, -1]$
- A particle starts moving from rest from a fixed point in a fixed direction. The distance s from the fixed point at a time t is given by $s = t^2 + at - b + 17$, where a, b are real numbers. If the particle comes to rest after 5 sec at a distance of $s = 25$ units from the fixed point, then values of a and b are respectively:
 (1) $10, -33$ (2) $-10, -33$
 (3) $-8, -33$ (4) $-10, 33$
- A stone is dropped into a quiet lake and waves moves in circles at a speed of 4cm/sec. At the instant, when the radius of the circular wave is 10 cm. How fast the enclosed area is increasing.
 (1) 40π (2) 80π
 (3) 100π (4) 60π
- The radius of a right circular cylinder increases at the rate of 0.1 cm/minute, and the height decrease at the rate of 0.2 cm/minute. The rate of change of the volume of the cylinder, in $\text{cm}^3/\text{minute}$, when the radius is 2cm and the height is 3 cm is
 (1) -2π (2) $-\frac{8\pi}{5}$
 (3) $-\frac{3\pi}{5}$ (4) $\frac{2\pi}{5}$
- Water runs into an inverted conical tent at the rate of 20 cu ft/min and leaks out at the rate of 5 cu ft/min . The height of the water is three times the radius of the water's surface. The radius of the water surface is increasing when the radius is 5ft, is
 (1) $\frac{1}{5\pi} \text{ ft/min}$ (2) $\frac{1}{10\pi} \text{ ft/min}$
 (3) $\frac{1}{15\pi} \text{ ft/min}$ (4) None of these
- The rate of change of the volume of a sphere w.r.t. its surface area, when the radius is 2 cm, is
 (1) 1 (2) 2
 (3) 3 (4) 4
- Let the equation of a curve be $x = a(\theta + \sin \theta)$, $y = a(1 - \cos \theta)$. If θ changes at a constant rate k then the rate of change of the slope of the tangent to the curve at $\theta = \frac{\pi}{3}$ is
 (1) $\frac{2k}{\sqrt{3}}$ (4) $\frac{k}{\sqrt{3}}$
 (3) k (4) $\frac{2k}{3}$
- The radius of a right circular cylinder increases at a constant rate. Its altitude is a linear function of the radius and increases three times as fast as radius. When the radius is 1 cm the altitude is 6cm. When the radius is 6cm, the volume is increasing at the rate of $1 \text{ cm}^3/\text{s}$. When the radius is 36 cm, the volume is increasing at a rate of $n \text{ cm}^3/\text{s}$. The value of ' n ' is equal to
 (1) 12
 (2) 22
 (3) 30
 (4) 33

- 10.** The point of the curve $y^2 = 2(x - 3)$ at which normal is parallel to the line $y - 2x + 1 = 0$ is:
- (1) $(5, 2)$ (2) $\left(-\frac{1}{2}, -2\right)$
- (3) $(5, -2)$ (4) $\left(\frac{3}{2}, 2\right)$
- 11.** The line of $x + y = 2$ is tangent to the curve $x^2 = 3 - 2y$ at its point.
- (1) $(1, 1)$ (2) $(-1, 1)$
- (3) $(\sqrt{3}, 0)$ (4) $(3, -3)$
- 12.** The slope of the normal to the curve $y^3 - xy - 8 = 0$ at the point $(0, 2)$ is equal to:
- (1) -3 (2) -6
- (3) 3 (4) 8
- 13.** The equation of the normal of $x^2 + y^2 - 2x + 4y - 5 = 0$ at $(2, 1)$ is:
- (1) $y = 3x - 5$ (2) $2y = 3x - 4$
- (3) $y = 3x + 4$ (4) $y = x + 1$
- 14.** The triangle formed by the tangent to the curve $f(x) = x^2 + bx - b$ at the point $(1, 1)$ and the co-ordinate axes, lies in the first quadrant. If its area is 2 then the value of b is
- (1) -1 (2) 3
- (3) -3 (4) 1
- 15.** If the slope of the curve $y = \frac{ax}{b-x}$ at the point $(1, 1)$ be 2, find the value of $a + b + 10$.
- 16.** If $y = 4x - 5$ is a tangent to the curve $y^2 = px^3 + q$ at $(2, 3)$, find the value of $(p - q - 4)$.



Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. (1)
2. (1)
3. (2)
4. (2)
5. (4)
6. (1)
7. (1)
8. (4)

9. (4)
10. (3)
11. (1)
12. (2)
13. (1)
14. (3)
15. (13)
16. (5)



PW Web/App - <https://smart.link/7wwosivoicgd4>

Library- <https://smart.link/sdfez8ejd80if>